Matthew Butrovich

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Summary

Database management systems (DBMS) expert with strong systems fundamentals spanning query engines, storage architectures, and operating systems (OS). Deep experience in analytical and transactional DBMS design: query execution and join algorithms, adaptive query execution and dynamic data pruning, storage management, and concurrency control. Complementary strengths in distributed systems, open table and file formats, and low-level performance engineering at the OS/DBMS boundary (CPU scheduling, eBPF, kernel networking, and observability). Active in the research and open source communities as a PMC member of Apache DataFusion and a frequent program committee member and reviewer, including for ACM SIGMOD and VLDB.

Professional Experience

2024 – **Apple** — Washington, DC

Senior Distributed Database Engineer

- Enhancing Apple's data lakehouse by contributing to multiple open source Apache projects: Arrow, DataFusion, DataFusion Comet, Iceberg, Parquet, and Spark.
- Brought Comet to production quality, reducing the cost of Spark pipelines by >50%.
- Enabled Comet's dynamic partition pruning (adaptive and non-adaptive query execution), which previously forced queries to fall back to slower Spark execution. Across 78 TPC-DS benchmark queries, the share of each query running natively rose from ~30–50% to ~95–97%, contributing to a >50% overall speedup at 1 TB.
- Designed and shipped Comet's native Iceberg scan, upstreaming >30 PRs to Iceberg.
- Drove a DataFusion sort-merge-join (SMJ) overhaul across ~12 PRs, making TPC-H >500x faster and near-unique LEFT/FULL joins >20x faster.
- Diagnosed and fixed a Comet busy-poll regression, dropping cluster CPU utilization from 30% to 8% on I/O-bound Iceberg workloads.
- Re-architected the JVM-to-native data path to true zero-copy using the Arrow C Stream Interface, eliminating a per-batch deep copy on every batch crossing the boundary.

2009 – 2017 **Western Digital Corporation** — Irvine, CA

Lead Technician V

- Led a team of 17 technicians verifying firmware for HDD and SSD storage devices.
- Promoted four times from initial role as Technician II while earning my B.Sc.

Education

2019 – 2024 **Carnegie Mellon University** — Pittsburgh, PA

Doctor of Philosophy (Ph.D.), Computer Science

- Advisor: Andy Pavlo
- Thesis: "On Embedding Database Management System Logic in Operating Systems via Restricted Programming Environments"

2017 – 2018 **Carnegie Mellon University** — Pittsburgh, PA

Master of Science (M.Sc.), Computer Science

2013 – 2017 **University of California, Irvine** — Irvine, CA

Bachelor of Science (B.Sc.), *cum laude*, Computer Science and Engineering

2010 – 2013 **Irvine Valley College** — Irvine, CA

Certificate of Achievement, Computer Science: Computer Languages

Selected Talks

2025 **CS 224P: Big Data Systems**
"Fusing Academic Research and Open Source Systems for Petabyte-Scale Analytics"

2025 **DC Systems**
"Putting Theory Into Practice"

2023 **15-445/645: Database Systems**
Lecture on join algorithms

2019 **15-445/645: Database Systems**
Lecture on two-phase locking concurrency control

Selected Publications

2025 **Matthew Butrovich**, Samuel Arch, Wan Shen Lim, William Zhang, Jignesh M. Patel, and Andrew Pavlo. "BPF-DB: A Kernel-Embedded Transactional Database Management System For eBPF Applications". *SIGMOD Conference.*

2023 **Matthew Butrovich**, Karthik Ramanathan, John Rollinson, Wan Shen Lim, William Zhang, Justine Sherry, and Andrew Pavlo. "Tigger: A Database Proxy That Bounces With User-Bypass". *Proc. VLDB Endow.*

2022 **Matthew Butrovich**, Wan Shen Lim, Lin Ma, John Rollinson, William Zhang, Yu Xia, and Andrew Pavlo. "Tastes Great! Less Filling! High Performance and Accurate Training Data Collection for Self-Driving Database Management Systems". *SIGMOD Conference.*

Service

To the Open Source Community

2026 – PMC Member, Apache DataFusion, *The Apache Software Foundation.*

2025 – Contributor, Apache Arrow, *The Apache Software Foundation.*

2025 – Contributor, Apache Iceberg, *The Apache Software Foundation.*

2025 – Contributor, Apache Parquet, *The Apache Software Foundation.*

2024 – Contributor, Apache Spark, *The Apache Software Foundation.*

2025 – 2026 Committer, Apache DataFusion, *The Apache Software Foundation.*

2024 – 2025 Contributor, Apache DataFusion, *The Apache Software Foundation.*

To the Profession

2026 Member, *ACM SIGMOD 2027 Program Committee.*

2026 Member, *VLDB 2027 Program Committee.*

2026 Member, *4th Workshop on eBPF and Kernel Extensions Program Committee.*

2025 Member, *ACM SIGMOD 2026 Industry Track Program Committee.*

2025 Member, *ACM SIGMOD 2026 Program Committee.*

2025 Reviewer, *ACM Transactions on Database Systems (TODS).*

2024 Member, *eBPF Summit 2024 Program Committee.*

2023 Member, *eBPF Summit 2023 Program Committee.*

2023 Member, *ACM SIGMOD 2023 Artifacts & Reproducibility Committee.*

Skills

Programming Languages: C, C++, eBPF, Java, Python, Rust, Scala
Development Tools: bpftrace, GDB/LLDB, Git, Intel VTune, perf